Software Modeling and Design Project

Topic: Property Rentals System

Group: A

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1. Overall description of the project

The Property Rentals System is a software application designed to facilitate the management of rental properties between property owners, tenants, and system administrators. The system provides a centralized platform where users can list rental properties, search for available accommodations, manage bookings, and handle rental agreements.

The platform allows property owners to publish information about their rental properties, including descriptions, prices, location, and availability. Tenants can browse available properties, view details, contact owners, and request bookings.The system also allows administrators to supervise the platform, manage user accounts, and ensure the smooth operation of the system.

The Property Rentals System aims to simplify the rental process by digitizing property listings, booking management, and communication between users.The application can be implemented as a web-based system or mobile application, allowing users to access rental services easily from anywhere with an internet connection.

2. Aim of the project

The main goal of the Property Rentals System is to:

- Simplify the process of renting properties
- Provide an organized platform for property listings
- Allow tenants to easily search and compare rental properties
- Help property owners manage their listings efficiently
- Automate booking and rental request processes
- Maintain records of users, properties, and rental transactions
- Improve transparency and communication between property owners and tenants
- Increase user satisfaction by providing an easy-to-use rental management platform

Overall, the system aims to make property rental management more efficient, reliable, and accessible.

3. User Levels

The system includes different categories of users called actors, each with specific roles and access permissions.

a) Primary actors (can also act as secondary actors depending on the use case)

1. User (Abstract actor)

These processes are done by all actors they just inherit it from the user.

- Register an account.
- Log into the account.
- Log out of the account.

2. Tenants (inherits from user)

Tenants are users who search for and rent properties.They can:

- Review and rate properties
- Search properties,view and request properties
- Communicate with property owners
- Make rental payments and view booking history

3. Property owner (inherits from user)

Property owners are users who list properties for rent. They can:

- Communicate with tenants
- Add,remove or edit property listings
- Recieve and manage rental requests
- View booking history

4. Administrator (inherits from user)

Administrators manage the whole system. They can:

- Monitor and control system operations
- Remove inappropriate content and manage disputes
- View reports and system statistics

5. Manager (inherits from user)

Managers oversee property operations, ensure policies are followed and coordinate between owners, tenants and administrators. They can:

- Monitor rental requests and approvals
- Generate management reports
- Handles complaints before they go to the administrator
- Approves large discounts or promotions

6. **Financiers (inherits from user)**

Financiers handle payments, transactions and financial reporting. They can:

- Monitors payment transactions and security
- Approve refunds and adjustments that can be made
- Communicates payment issues with tenants or owners
- Generate financial reports

7. **Human Resources (HR) (inherits from user)**

HR handles user support, communication, and dispute management within the system. Responsibilities include:

- Managing complaints and disputes
- Sending notifications and reminders
- Monitoring user satisfaction and feedback
- Maintaining records of issues and resolutions

b) External Actors

Payment System
- Processes rental payments and transactions

Notification System
- Sends notifications and alerts to users regarding bookings, confirmations, or updates

4. General Requirements

Functional Requirements

- Users should be able to register and log into the system
- Property owners should be able to add, edit, and remove property listings
- Tenants should be able to search properties based on location, price, and type
- Tenants should be able to send rental requests or booking requests
- Property owners should be able to accept or reject rental requests
- The system should store property information, booking history, and user data
- The system should allow communication between tenants and property owners
- Administrators should be able to manage users and listings
- The system should allow payment processing for rental transactions

Non-Functional Requirements

- Performance: The system should support multiple users accessing the system simultaneously without performance issues.
- Security: User information and payment data must be securely stored and protected through authentication and authorization mechanisms.
- Usability: The user interface should be simple, intuitive, and easy to use for all types of users.

- Reliability:The system should operate with minimal downtime and maintain data integrity.
- Scalability:The system should be able to scale as the number of users and property listings increases.
- Maintainability:The system should be designed in a way that allows easy maintenance and future updates.
- Availability: The system should be accessible to users when they need it.
- Externals: The system must deliver high performance, secure data handling, intuitive usability, reliable operation, scalable growth, easy maintainability, and consistent availability to meet user and business needs.

5. Scope of the System

Includes

- Property listing and management
- Property search functionality
- Rental request management
- User account management
- Admin management panel
- Payment handling

Does Not Include

- Integration with external real estate agencies
- Legal document generation for rental contracts
- AI-based property recommendation systems (initial version)

6. Assumptions and Constraints

Assumptions

- Users will have internet access
- Property owners will provide accurate property information
- Tenants will use the system to search and request properties
- The system will be accessed through web or mobile devices

Constraints

- Limited development time for the project
- Budget limitations for advanced system features
- Dependence on third-party services for payment processing
- Internet connectivity required for system access

7. Simple System Overview

The tenant accesses the system and searches for available rental properties based on preferences such as location, price, or property type. The system displays available properties with detailed information. The tenant selects a property and sends a rental request to the property owner. The owner reviews the request and either accepts or rejects it. Once the request is approved, the tenant can proceed with payment through the system. The administrator monitors the system, manages users and listings, and ensures that the platform operates properly.